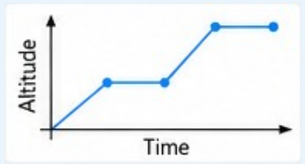
CALM MIL Schematic

MISSION & ENVIRONMENT INPUTS

Mission profile



Altitude, velocity and flight phases

Atmosphere & wind



Wind, density, temperature and gravity

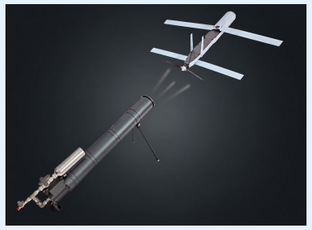
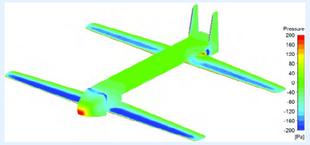
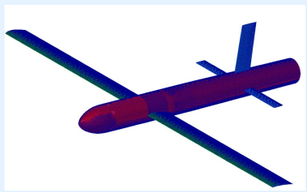
Initial conditions

Position, velocity, attitude and rates

Launcher configuration

Canister geometry, ejection parameters, launch attitude

AIRCRAFT PLANT MODELS (SIMCENTER)

Launcher / Canister System (Ejection Phase Only) 	**Model description** Ejection dynamics; separation sequence; initial conditions to free flight.
Aerodynamics (3D-derived / ROM) 	**Model description** Aerodynamic force/moment coefficient maps versus Mach, angle of attack, sideslip, control deflection and wing configuration. CFD-derived; optional ROM.
Propulsion – Electric Motor, Propeller & Battery	**Model description** Propeller, motor, ESC and LiPo battery dynamics: SOC, OCV, voltage, internal resistance and power.
Structures / Flexible Dynamics 	**Model description** FEA-derived loads, modal response, flexibility, wing deployment and locking; ROM where necessary.
Actuator Models (in Plant) 	**Model description** Control-surface servos and wing deployment actuators; physical rate/position limits, dynamics and delays.
6-DOF Rigid-Body Dynamics 	**Model description** Force and moment balance, translational/rotational motion, mass properties and configuration-dependent inertia.

INTERFACES & INTEGRATION

Signal buses

Plant states → FCS; FCS commands → plant; actuator, engine, launcher and environment signals.

Simulation environment

Amesim, Simulink solver settings, scenarios, parameters and logging.

FLIGHT CONTROL SYSTEM (MATHWORKS)

Existing FCS Model – Supplied

Existing customer Simulink logic model. Internal implementation is not specified or reproduced.

External input bus

Vehicle states, rates, air data and relevant subsystem feedback.

External output bus

Control surface and propulsion commands, mode/status outputs.

MIL connection

Closed-loop exchange via agreed Simulink interfaces.

OUTPUTS & APPLICATIONS

Vehicle states

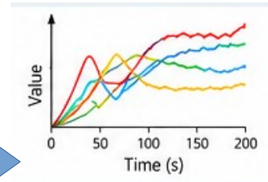
Position, velocity, attitude, rates, accelerations; launcher/deployment states.

Performance metrics

Flight path, tracking error, stability, energy, control effort and loads.

Post-processing

Time histories; comparisons with test logs; V&V and what-if studies.



- Time histories of variables
- Comparison with test data
- V&V reports
- What if studies?